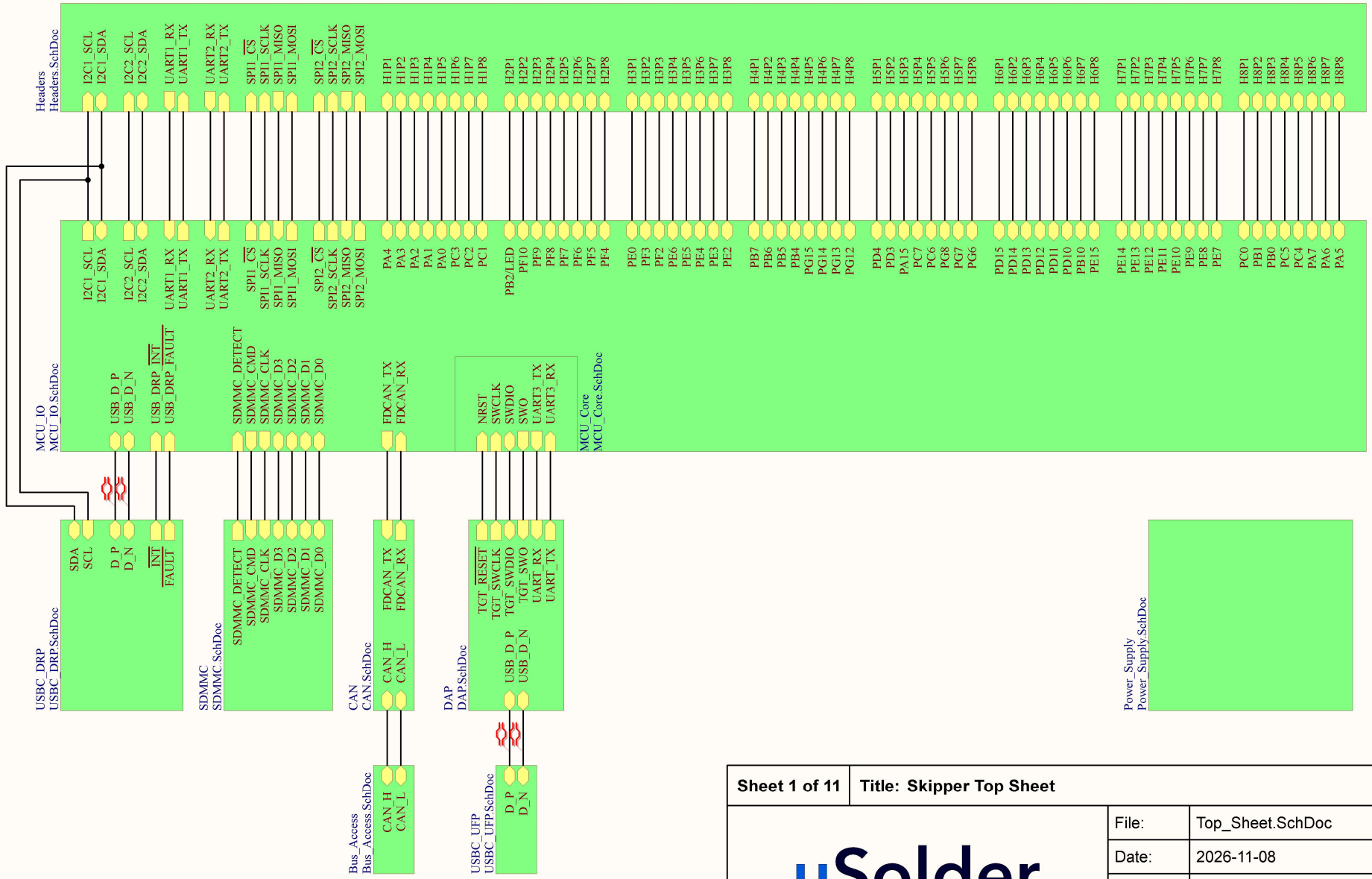
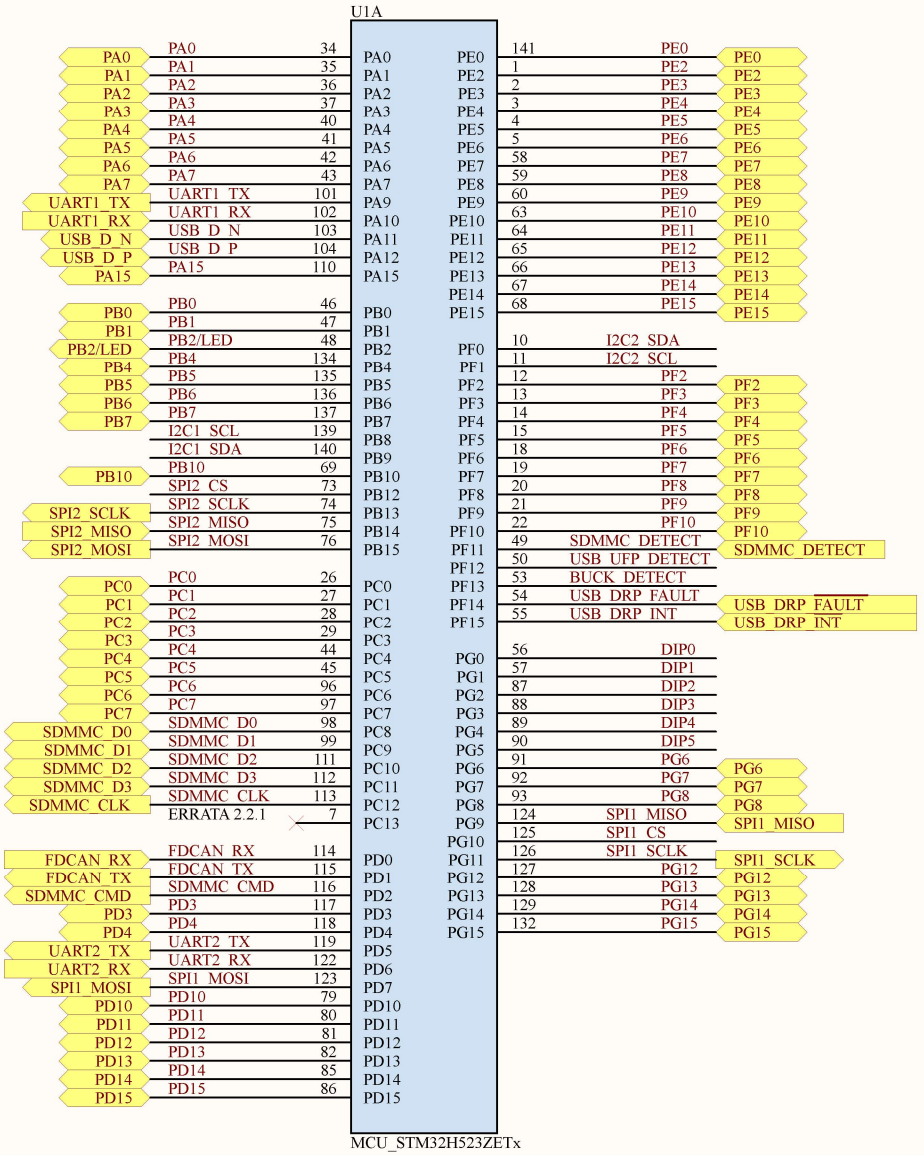
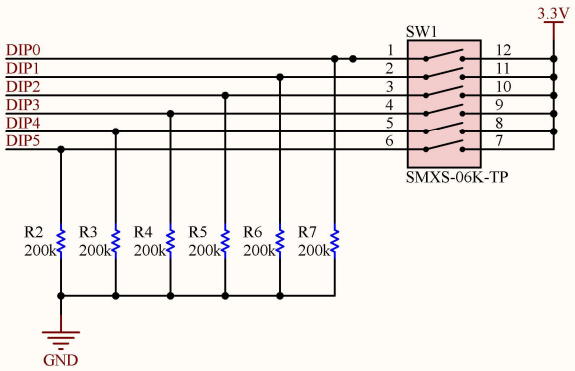


Sheet 1 of 11	Title: Skipper Top Sheet
uSolder	File: Top_Sheet.SchDoc
	Date: 2026-11-08
	Author: Jack Herron
	Project: Skipper.PrjPcb

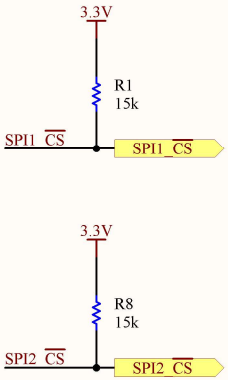
STM32H523 Ports



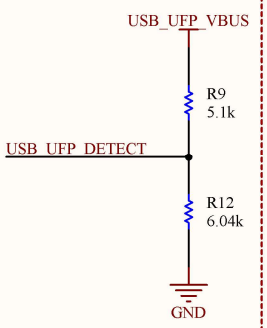
User 6-bit DIP Switch



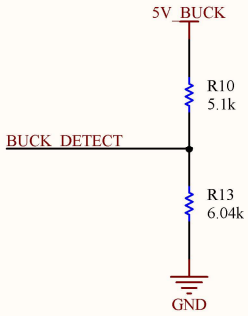
SPI CS Pull-ups



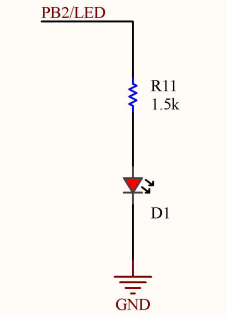
USB UFP Detection



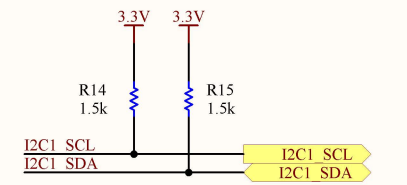
Buck 5V Detection



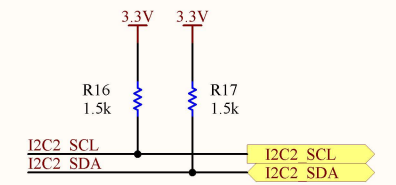
User LED



I2C1 Pull-ups



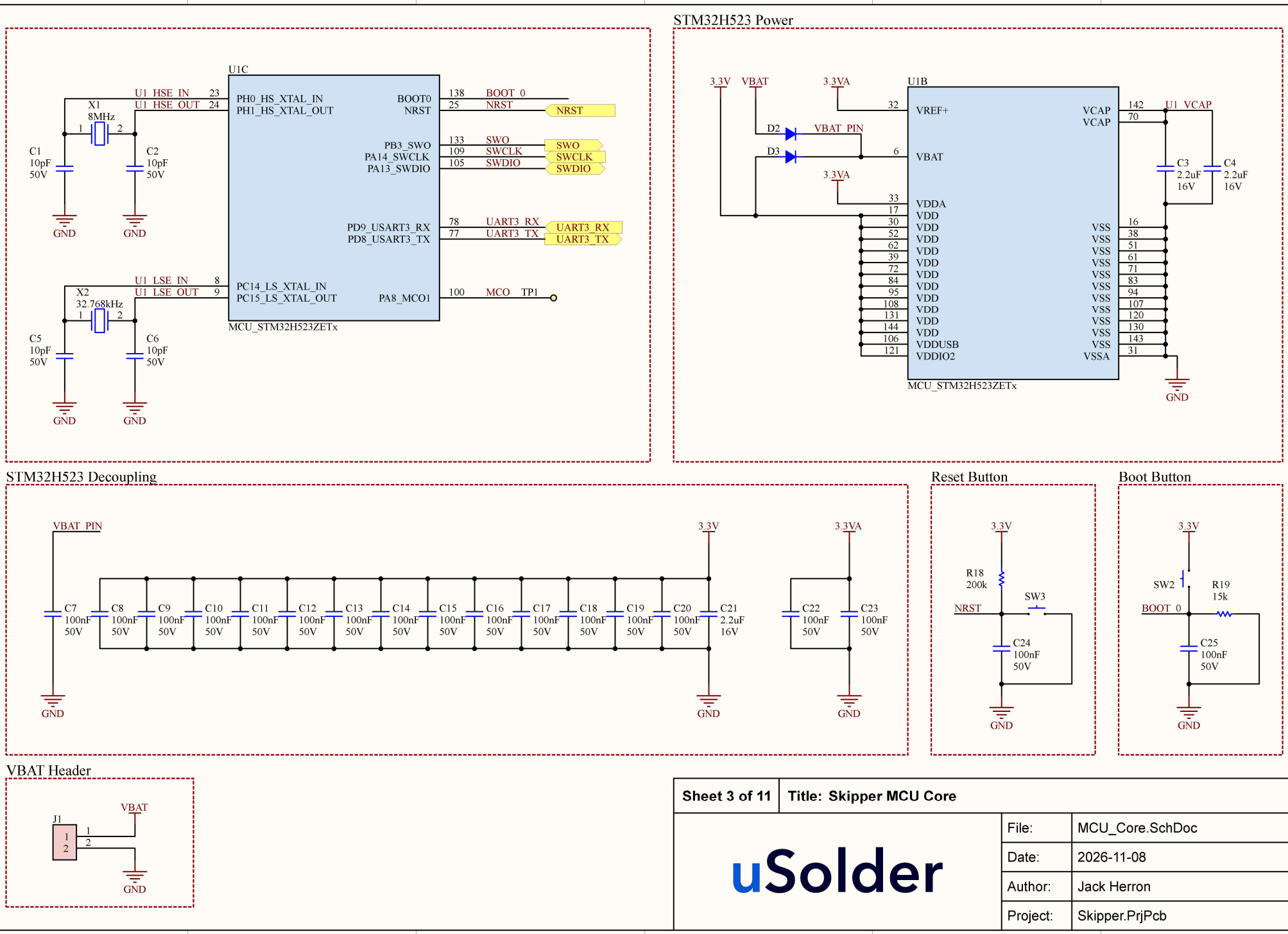
I2C2 Pull-ups

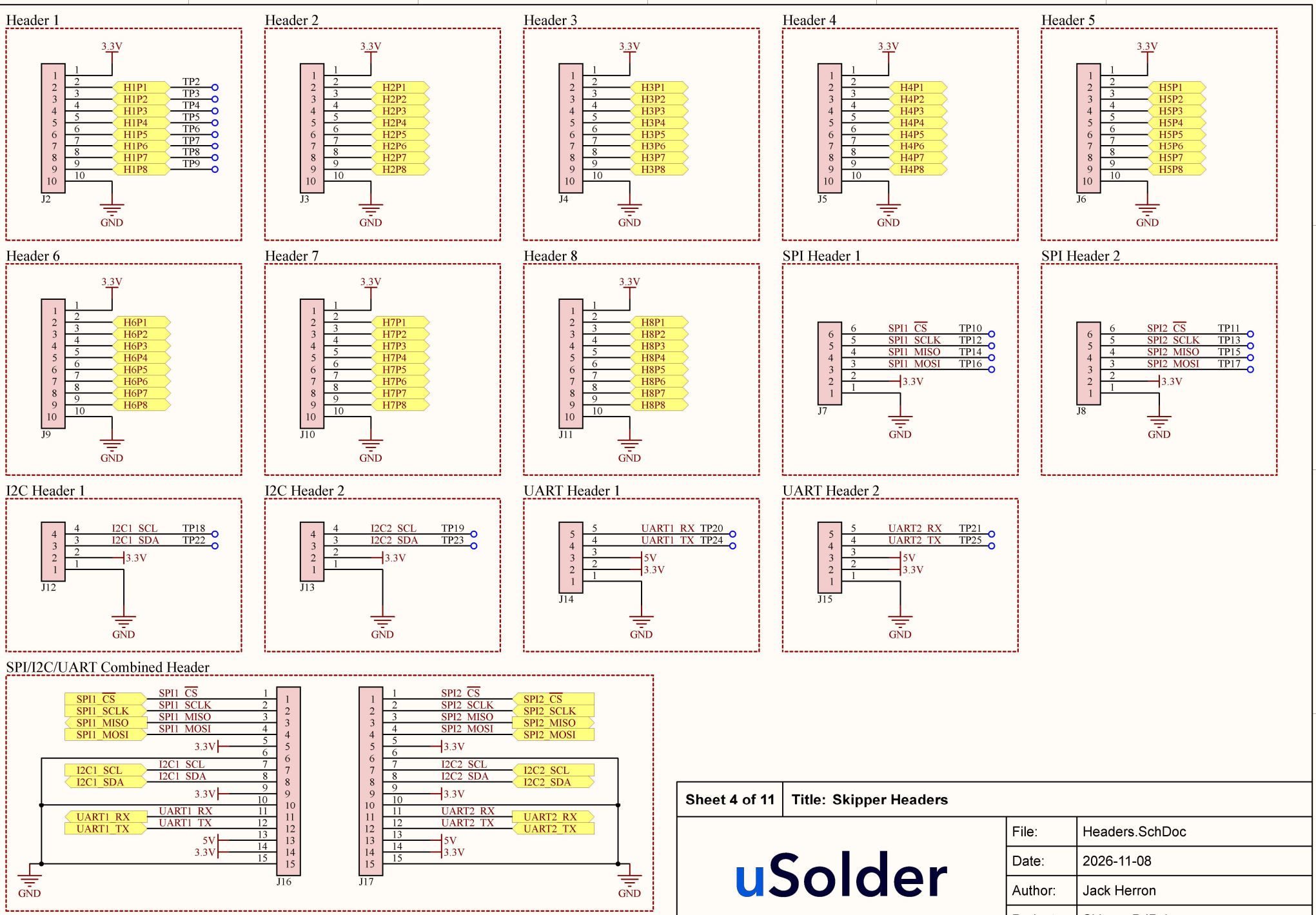


Sheet 2 of 11	**Title: Skipper MCU IO**

uSolder

File:	MCU_IO.SchDoc
Date:	2026-11-08
Author:	Jack Herron
Project:	Skipper.PrjPcb


U1C
U1_HSE_IN
U1_HSE_OUT
X1
8MHz
C1
10pF
50V
C2
10pF
50V
GND
PH0_HS_XTAL_IN
PH1_HS_XTAL_OUT
BOOT0
NRST
PB3_SWO
PA14_SWCLK
PA13_SWDIO
PD9_USART3_RX
PD8_USART3_TX
PC14_LS_XTAL_IN
PC15_LS_XTAL_OUT
PA8_MCO1
BOOT_0
SWO
SWCLK
SWDIO
UART3_RX
UART3_TX
MCO
TP1
U1_LSE_IN
U1_LSE_OUT
X2
32.768kHz
C5
C6
MCU_STM32H523ZETx
STM32H523 Power
3.3V
VBAT
3.3VA
U1B
D2
D3
VBAT_PIN
VREF+
VBAT
VDDA
VDD
VSS
VCAP
VDDUSB
VDDIO2
VSSA
U1_VCAP
C3
2.2uF
16V
C4
2.2uF
16V
MCU_STM32H523ZETx
STM32H523 Decoupling
C7
C8
C9
C10
C11
C12
C13
C14
C15
C16
C17
C18
C19
C20
C21
2.2uF
16V
C22
C23
100nF
50V
Reset Button
R18
200k
SW3
NRST
C24
100nF
50V
Boot Button
SW2
R19
15k
BOOT_0
C25
100nF
50V
VBAT Header
J1
VBAT
Sheet 3 of 11
Title: Skipper MCU Core
uSolder
File: MCU_Core.SchDoc
Date: 2026-11-08
Author: Jack Herron
Project: Skipper.PrjPcb

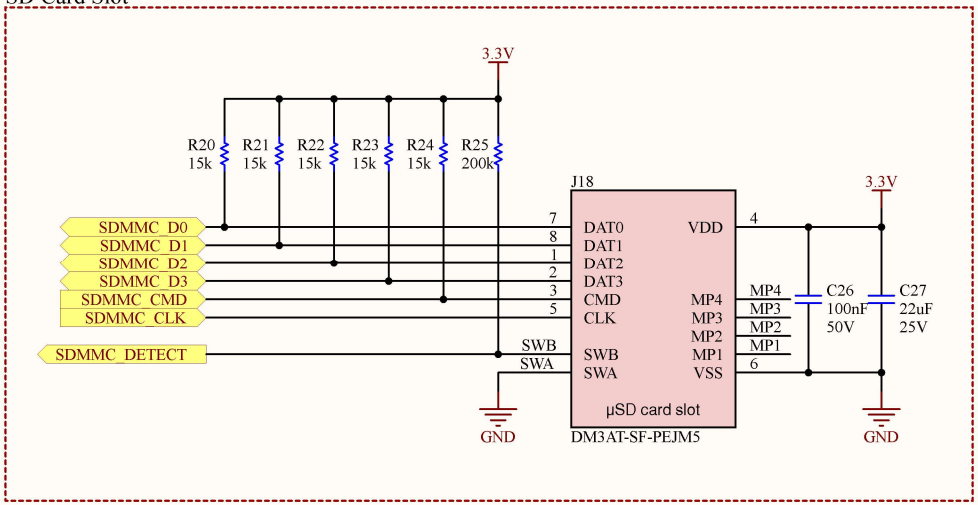


Sheet 4 of 11	**Title: Skipper Headers**

uSolder

File:	Headers.SchDoc
Date:	2026-11-08
Author:	Jack Herron
Project:	Skipper.PrjPcb

SD Card Slot

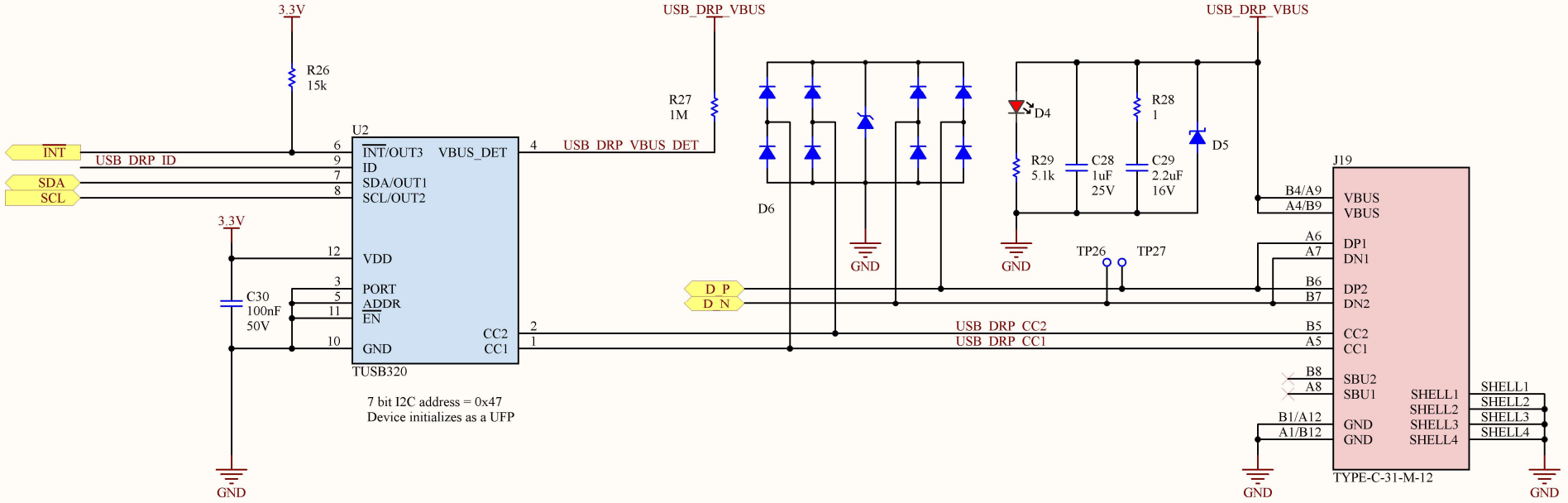


Sheet 5 of 11	**Title: Skipper SD Card Slot**

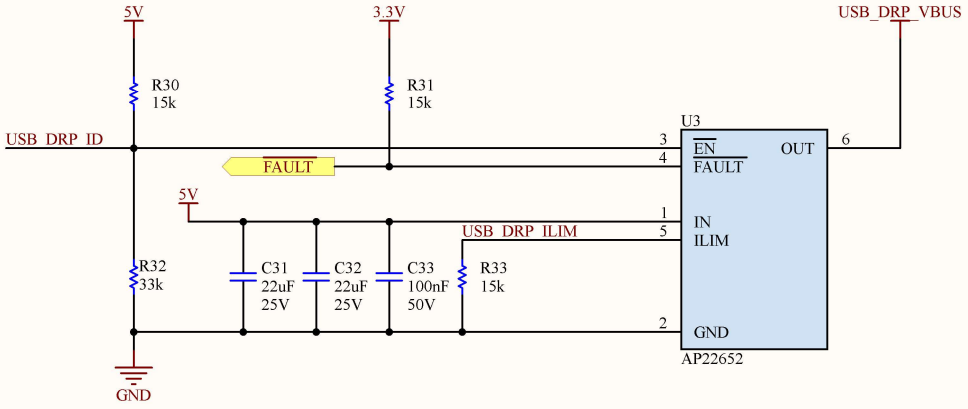
uSolder

File:	SDMMC.SchDoc
Date:	2026-11-08
Author:	Jack Herron
Project:	Skipper.PrjPcb

USBC DRP

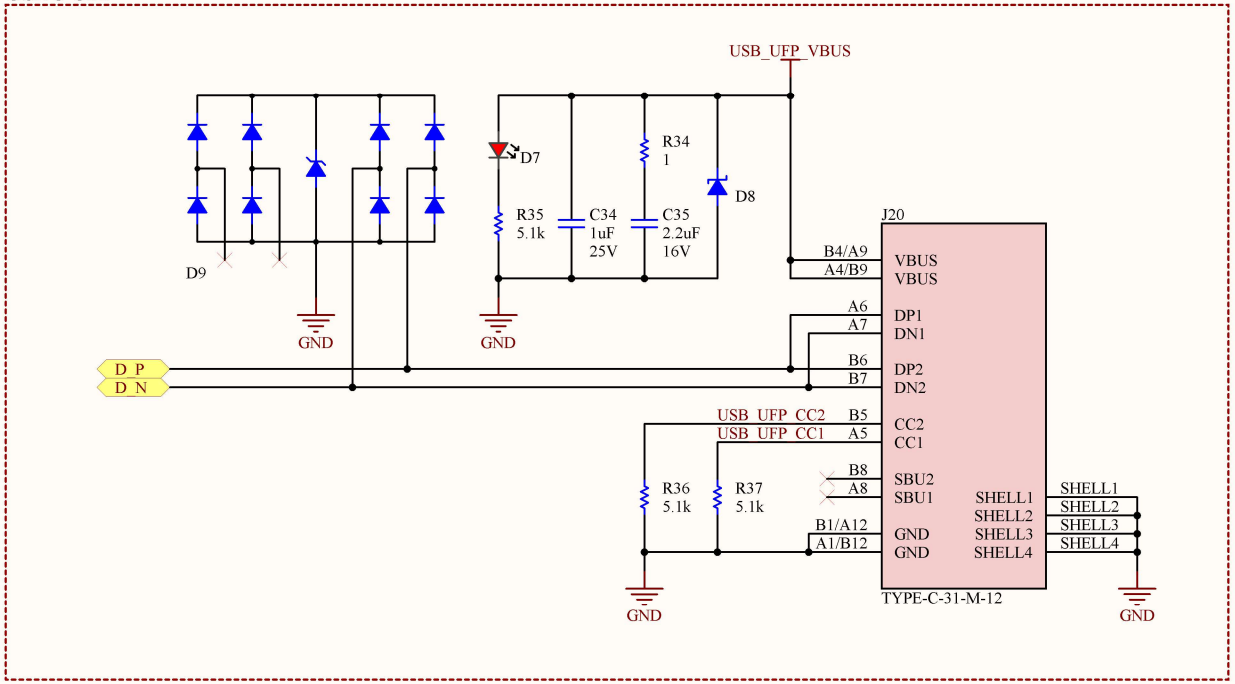


USB Host Power Switch



Sheet 6 of 11	**Title: Skipper USBC DRP**
uSolder	File: USBC_DRP.SchDoc
	Date: 2026-11-08
	Author: Jack Herron
	Project: Skipper.PrjPcb

USBC UFP

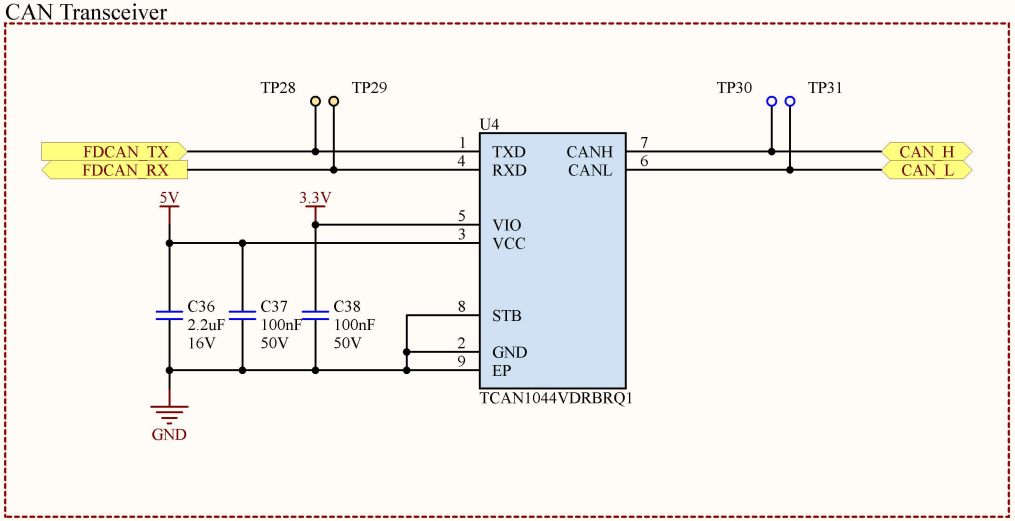


Sheet 7 of 11	**Title: Skipper USBC UFP**

uSolder

File:	USBC_UFP.SchDoc
Date:	2026-11-08
Author:	Jack Herron
Project:	Skipper.PrjPcb

CAN Transceiver

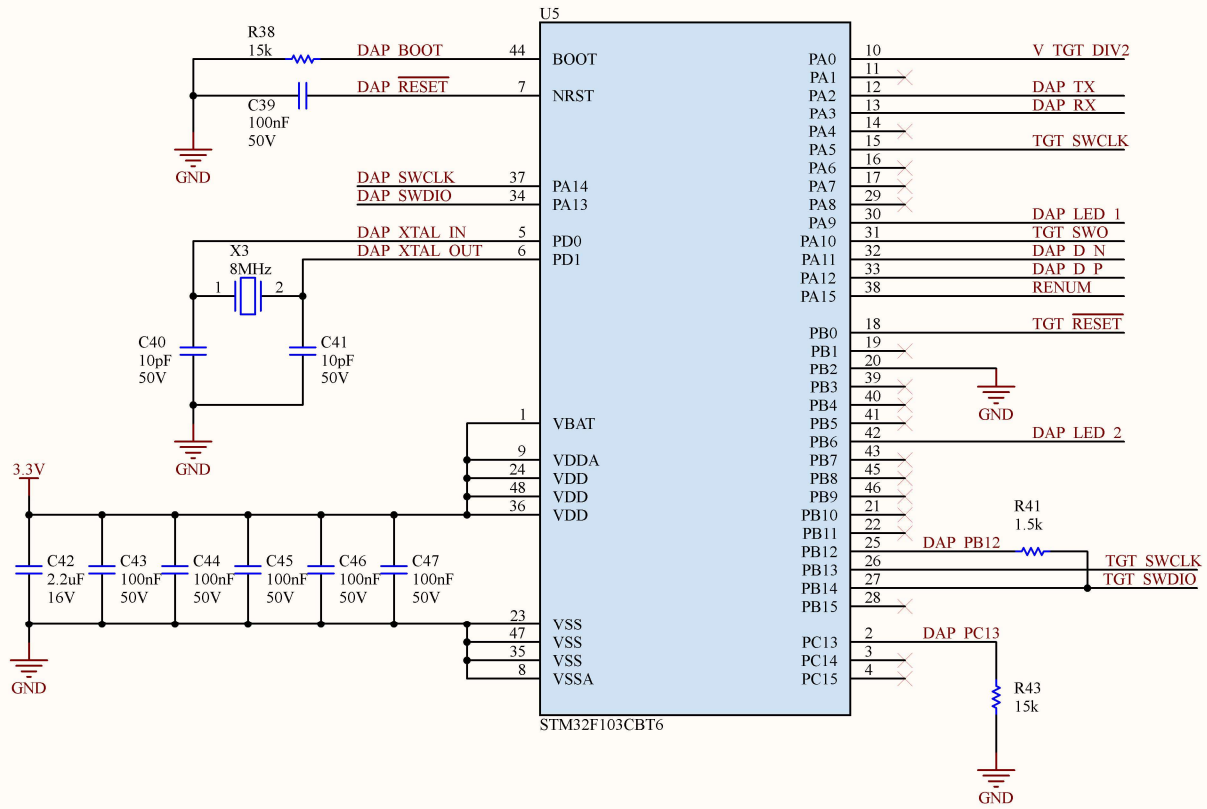


Sheet 8 of 11	**Title: Skipper CAN Transceiver**

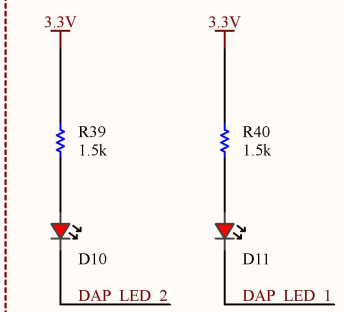
uSolder

File:	CAN.SchDoc
Date:	2026-11-08
Author:	Jack Herron
Project:	Skipper.PrjPcb

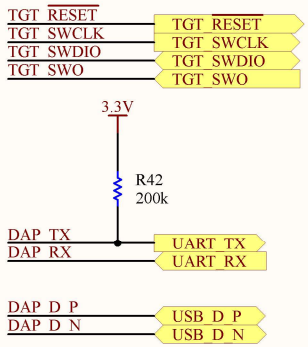
STM32F103 HIC



DAP LEDs



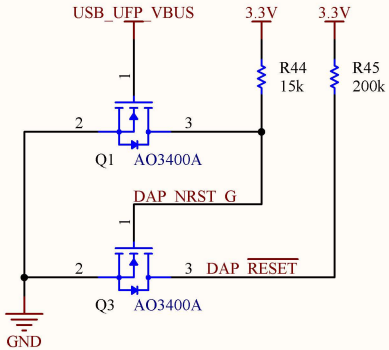
IO



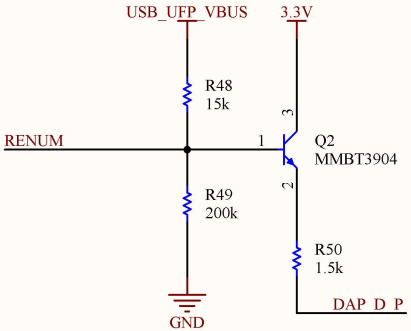
DAP Programming Header



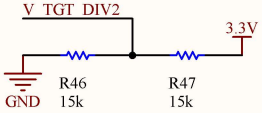
DAP Reset Circuit



USB RENUM Control



DAP Target Voltage Divider

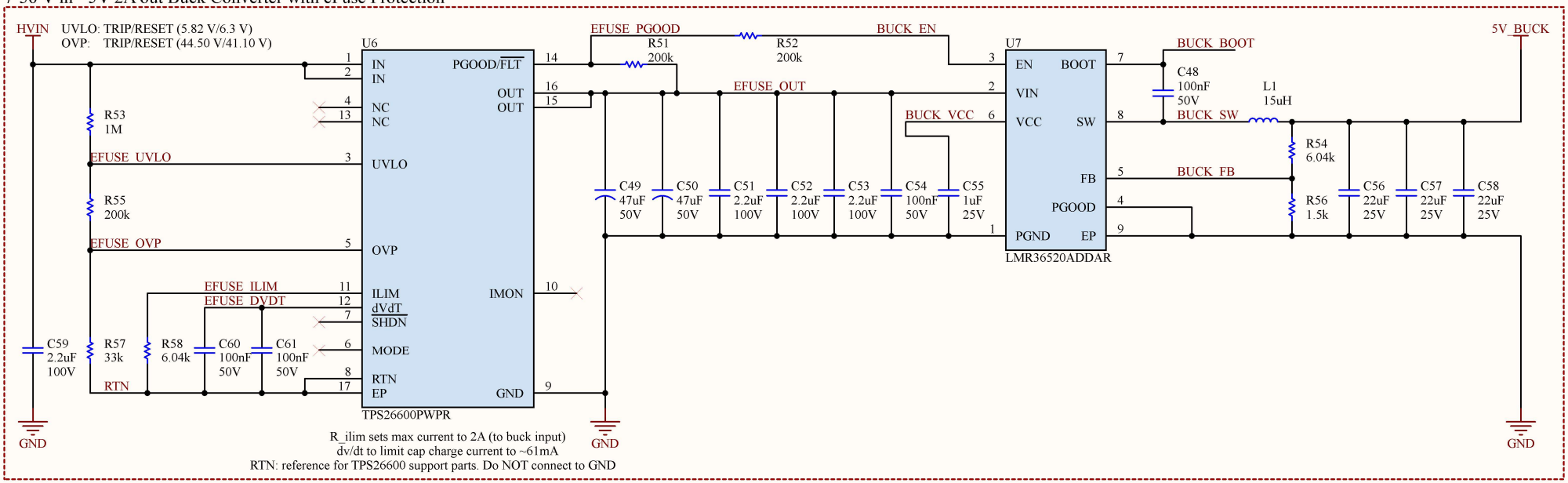


Sheet 9 of 11	**Title: Skipper DAP**

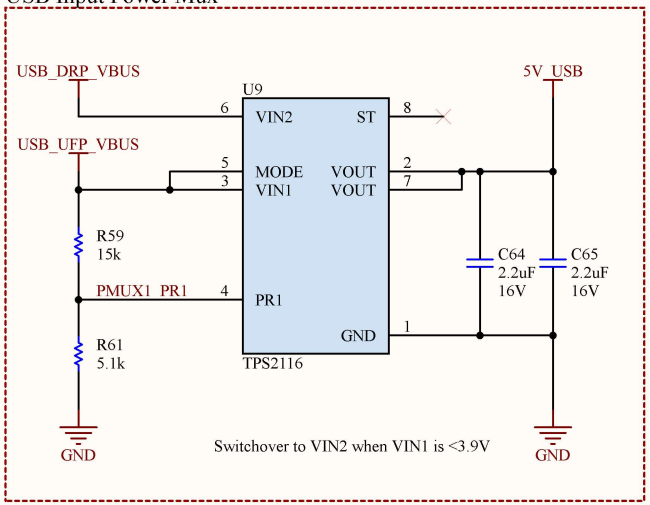
uSolder

File:	DAP.SchDoc
Date:	2026-11-08
Author:	Jack Herron
Project:	Skipper.PrjPcb

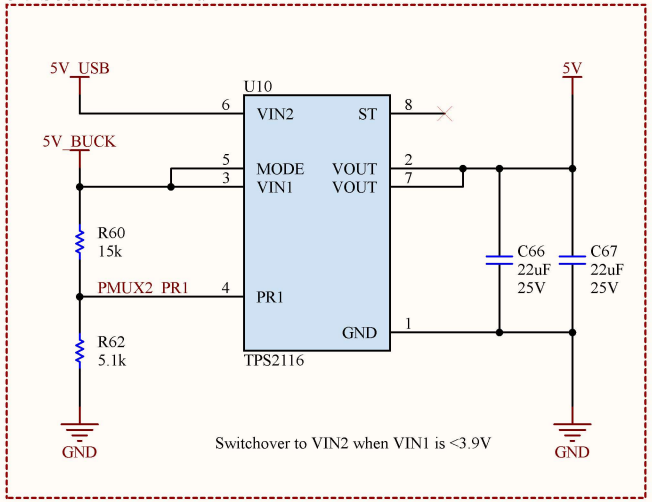
7-36 V in - 5V 2A out Buck Converter with eFuse Protection



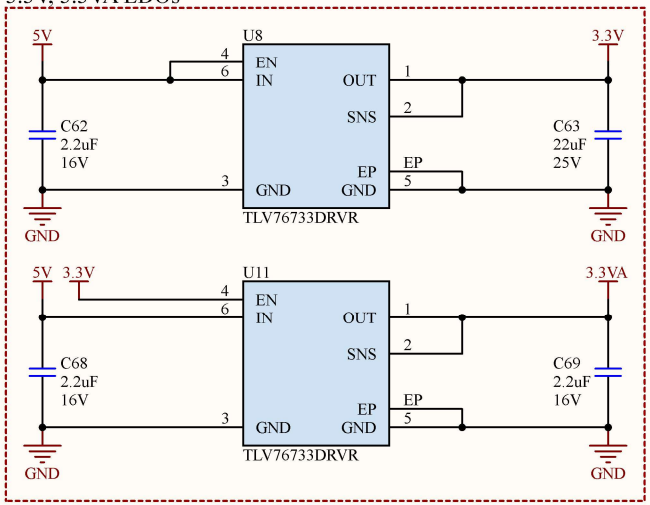
USB Input Power Mux



5V Source Power Mux

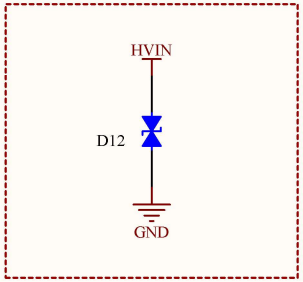


3.3V, 3.3VA LDOs

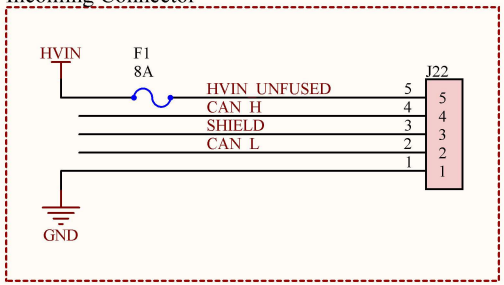


Sheet 10 of 11	Title: Skipper Power Supply
uSolder	File: Power_Supply.SchDoc
	Date: 2026-11-08
	Author: Jack Herron
	Project: Skipper.PrjPcb

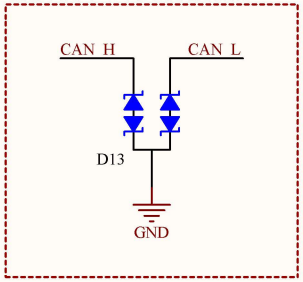
HVIN 36V TVS Protection



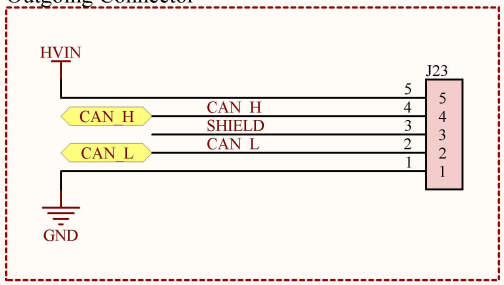
Incoming Connector



CAN 36V TVS Protection



Outgoing Connector



Sheet 11 of 11	**Title: Skipper Bus Connection**
uSolder	File: Bus_Access.SchDoc
	Date: 2026-11-08
	Author: Jack Herron
	Project: Skipper.PrjPcb